

My CAD Designs and Engineering Projects

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Boston University – Mechanical Engineering with Concentrations in
Manufacturing and Technology Innovation

1. Morphable Biorobotics Lab
2. Box Measuring the Temperature and Alerting the Surrounding if the
Temperature Is Out Of the Range
3. Rocket Propulsion Group
4. Mars Rover Club Robotic Arm
5. BAJA Club Off-Road Automotive Frame Design
6. Additional SolidWorks Designs
7. Retro Passage Bell Project

CAD Design and Engineering Project 1 – Morphable **Biorobotics Lab**

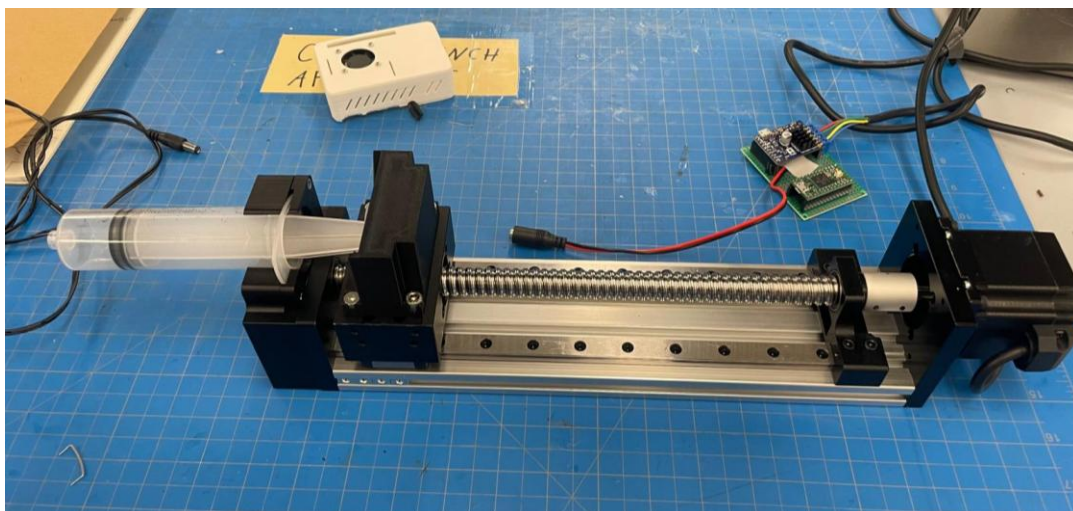
For more than a year, I work as a researcher in Professor Ranzani's Morphable Biorobotics Lab at Boston University, where I gain hands-on experience in the manufacturing and development of soft-robotic systems for surgical applications. I collaborate with masters and PhD students as well as professors which makes this experience a rewarding journey.

This lab develops a robotic platform for surgical procedures inside the beating heart to replace the tricuspid valve. I am working with Leonardo Zamora, a PhD student focusing on the development of the soft robotic platform. This project has immersed me in the realities of designing for dynamic, anatomically constrained environments in which precision and biocompatibility are paramount.

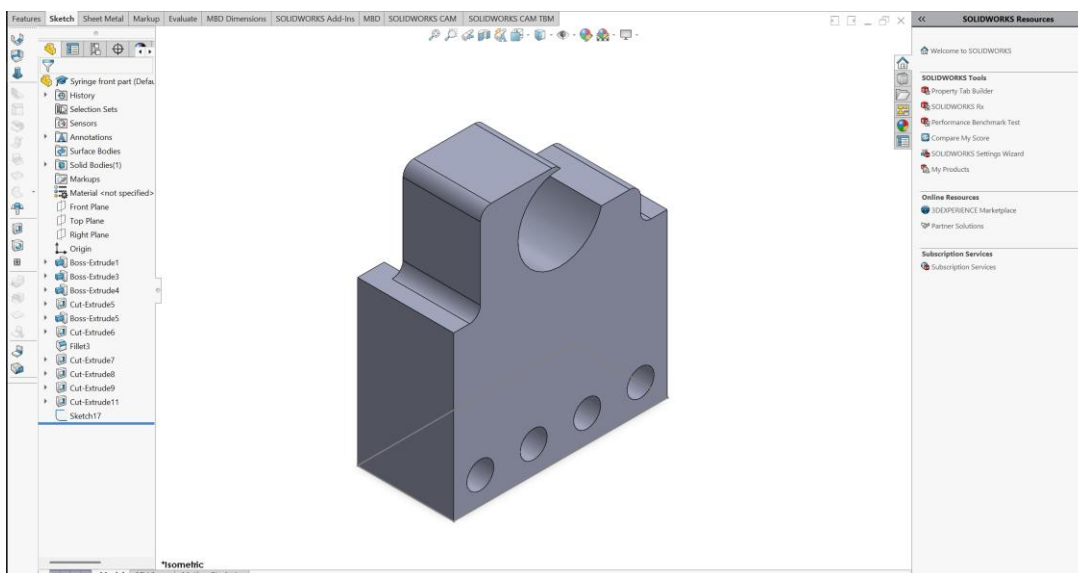
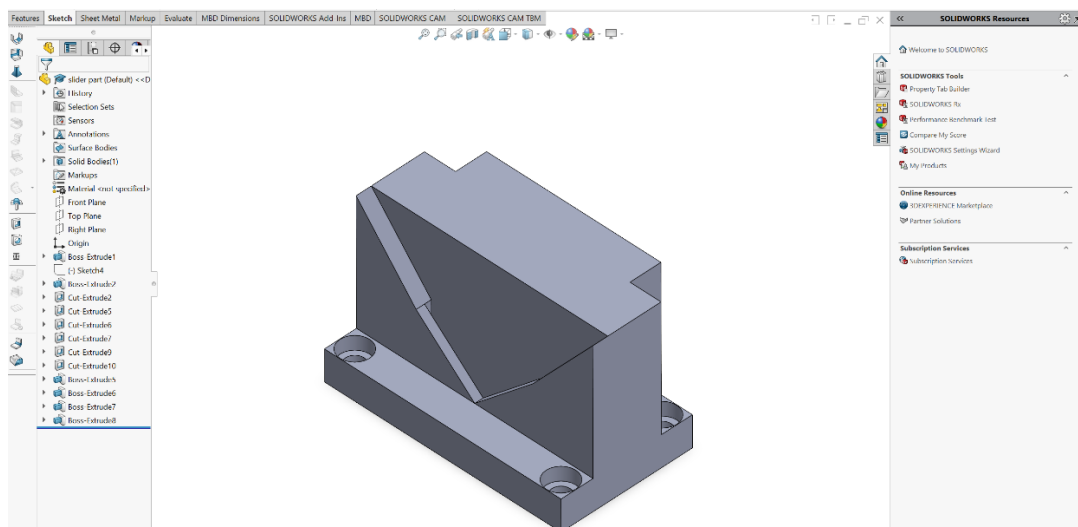
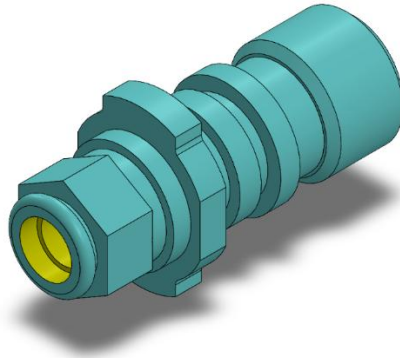
Day to day, I translate clinical and system constraints into mechanical requirements, iterate CAD concepts, and bring designs to life through rapid prototyping workflows. Working directly with soft materials, I prototype components, evaluate their mechanical behavior under physiologically relevant motions, and refine designs to achieve repeatable actuation and controlled compliance.

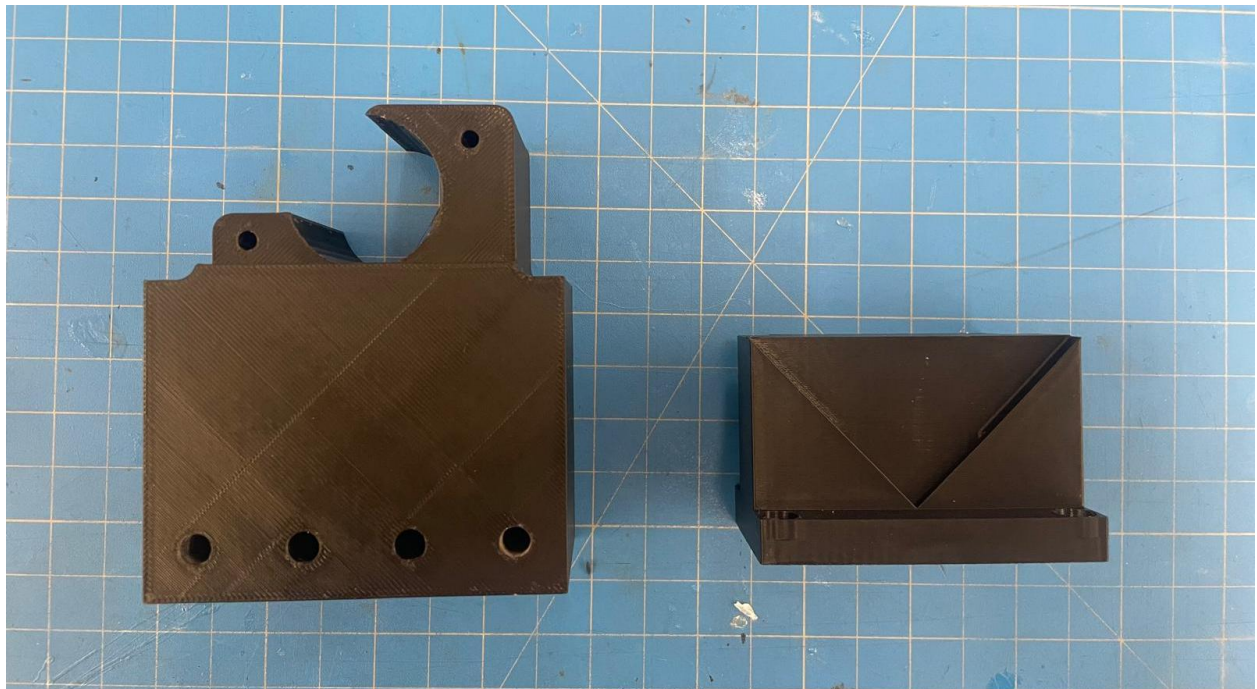
Thus, this experience has deepened my skills in CAD modeling using AutoCAD, Solidworks and NX, and rapid prototyping like 3D printing by using Bambu Studio.

Below, you can see the syringe pump system for the robotic platform. This is powered by a motor which enables the automation of the syringe pump movement forward and backwards. I CAD designed and 3D printed all the components.



Below you can also see the some CAD component designs separately before I printed and assembled them.





In addition, I designed a box which can hold the cables and small hoses in place. This part is very important for fluidic actuation which can be seen in part iii below.

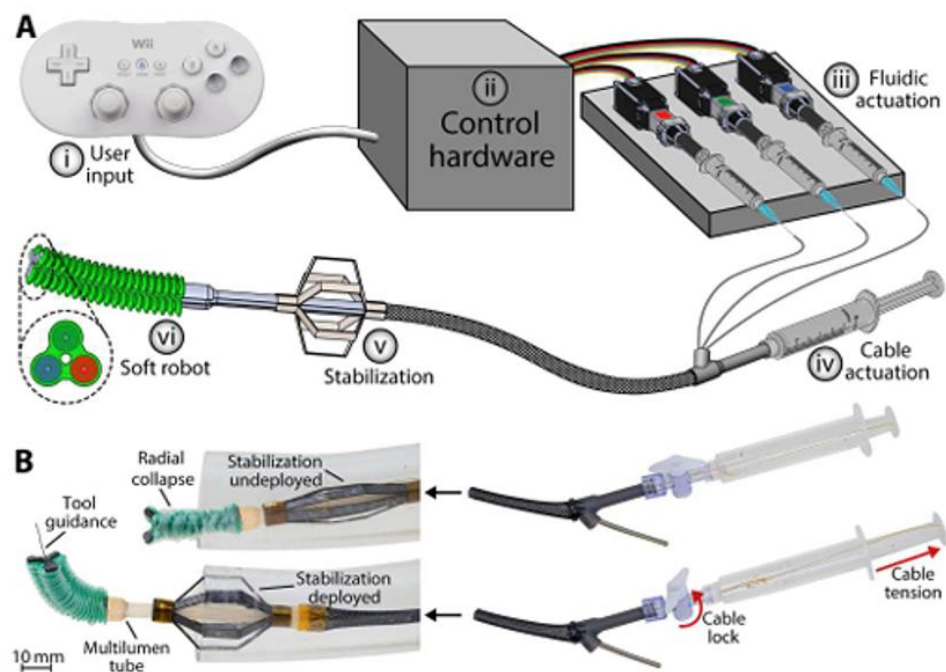
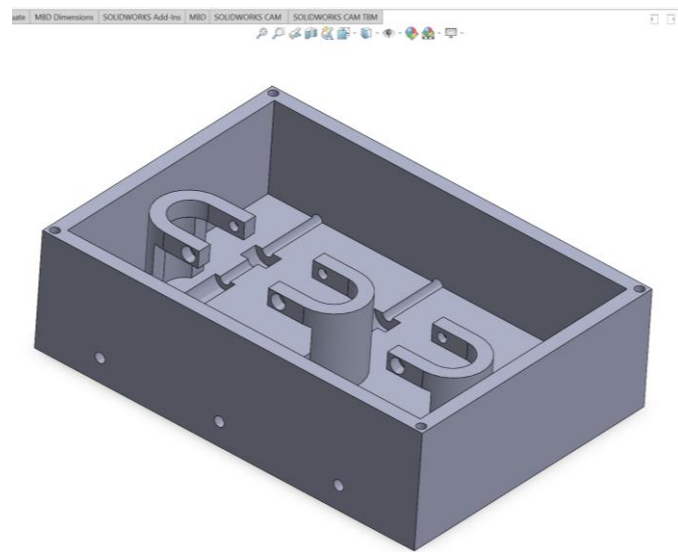
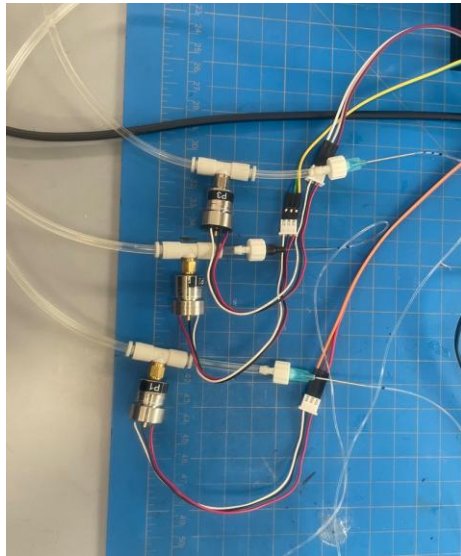
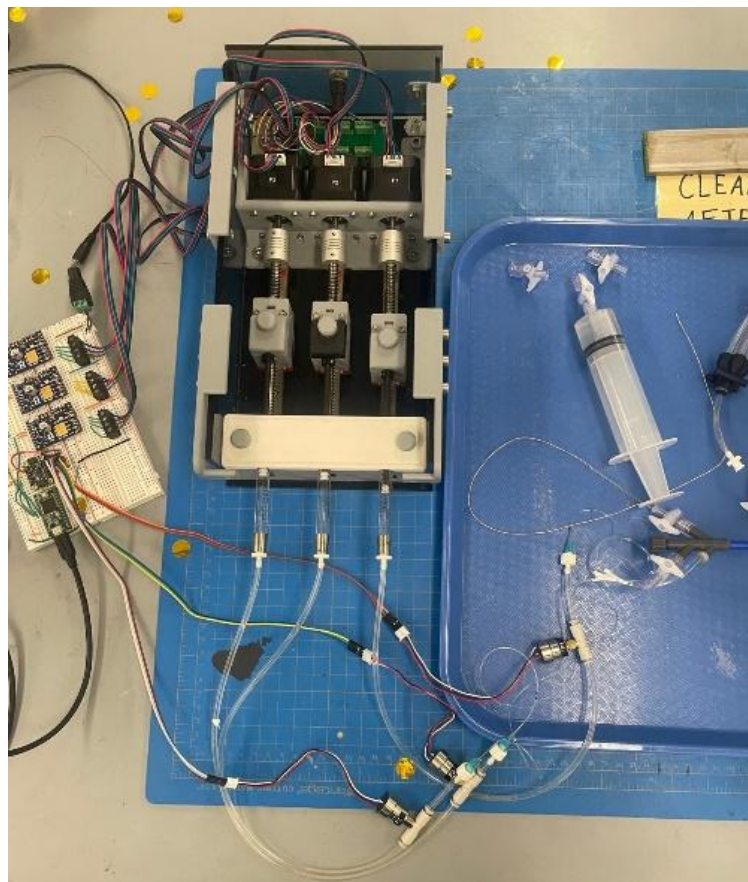


Fig. 3. Integrated robotic platform overview. (A) The device is operated with a

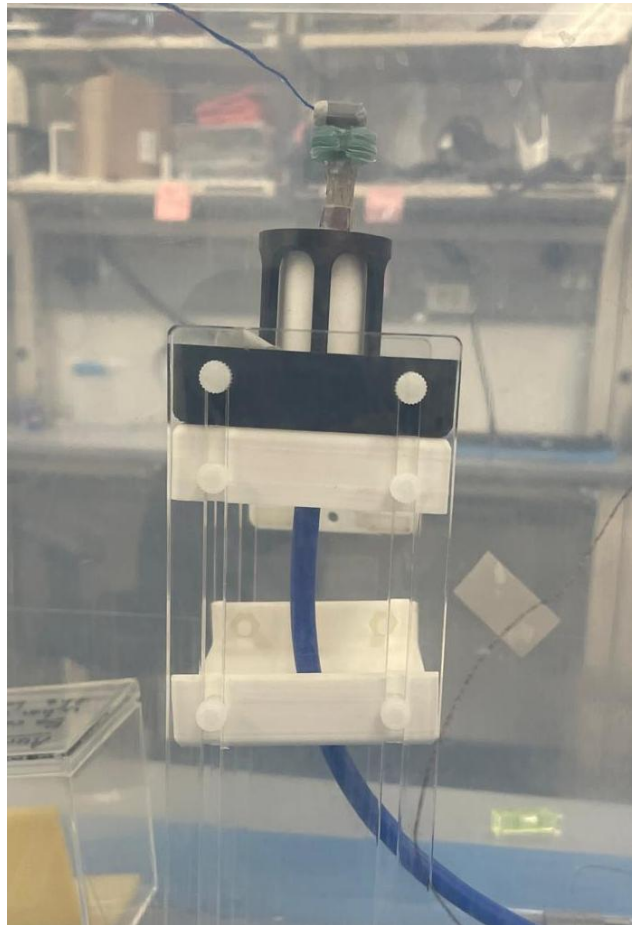
So, I CAD designed the box below to hold important fluidic actuation components in place.



Now, I am working on assembling and circuiting different components which forms much of the integrated robotic platform control hardware, fluidic and cable actuation.



These components will be connected to the soft robot shown below. The user input from the console would enable the surgeon to control the soft robot inside the beating heart to replace the tricuspid valve.

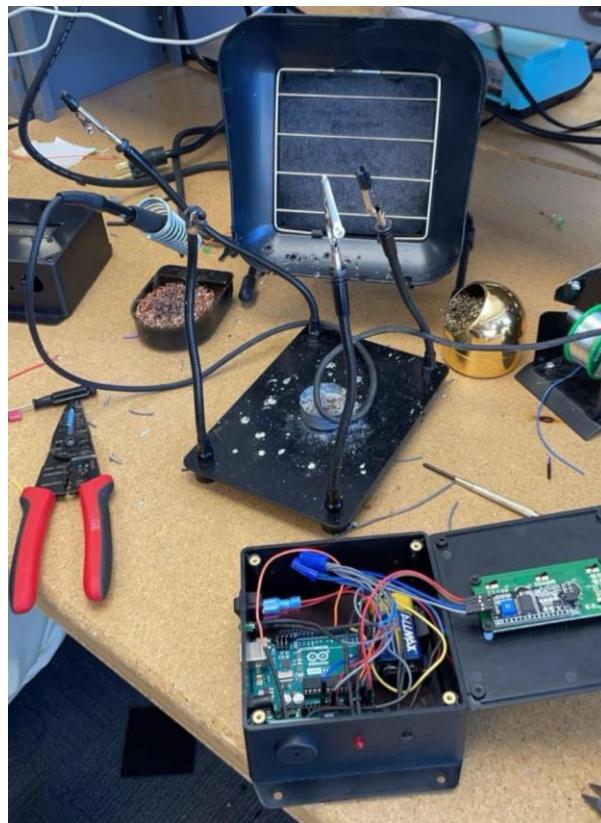


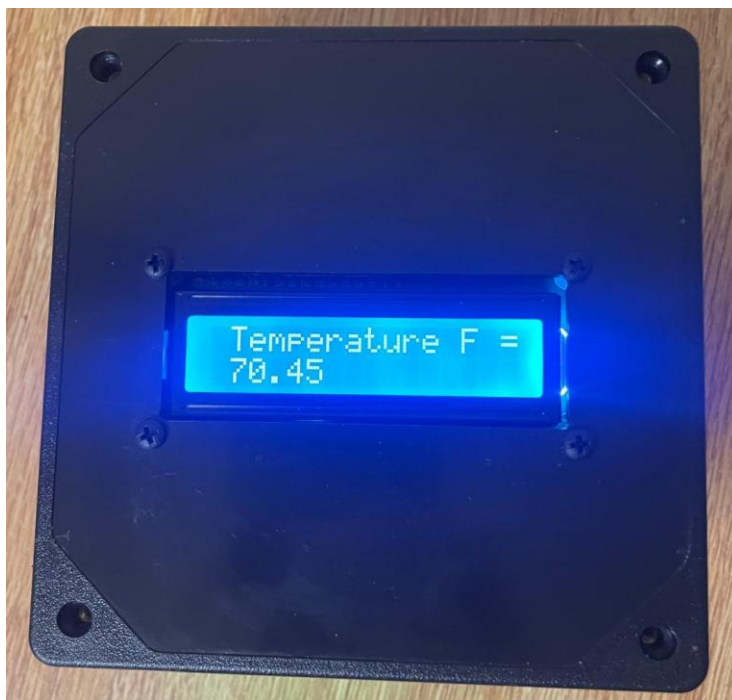
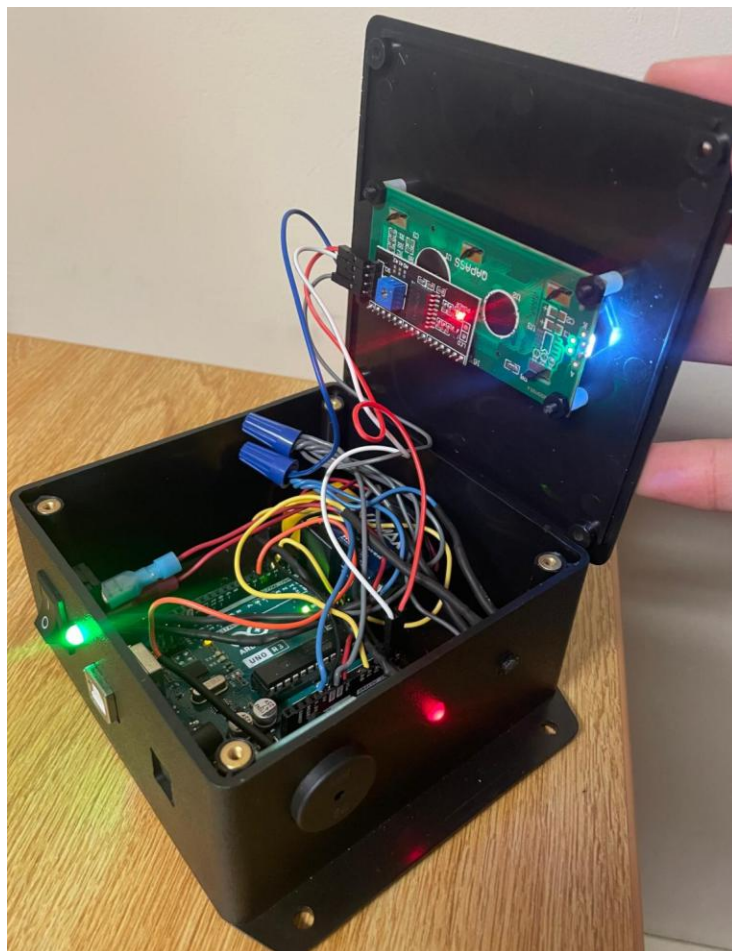
CAD Design and Engineering Project 2 – Box Measuring the Temperature and Alerting the Surrounding if the Temperature Is Out Of the Range

In this project, I used a set of design-oriented engineering skills through a hands-on experience using engineering tools for design and prototyping. This project benefited me to develop my both hardware and software skills. I also improved myself on designing and wiring circuits, prototyping, instrumentation, 3D visualization and building structures.

Building substructures and assembling them to develop a functional final product which is used to measure the temperature inside the room was a rewarding journey. Also, the circuitry and software usage enables this box to warn the surroundings through alerting with the red light. This makes the design an important final product which can be used in different settings and industries.

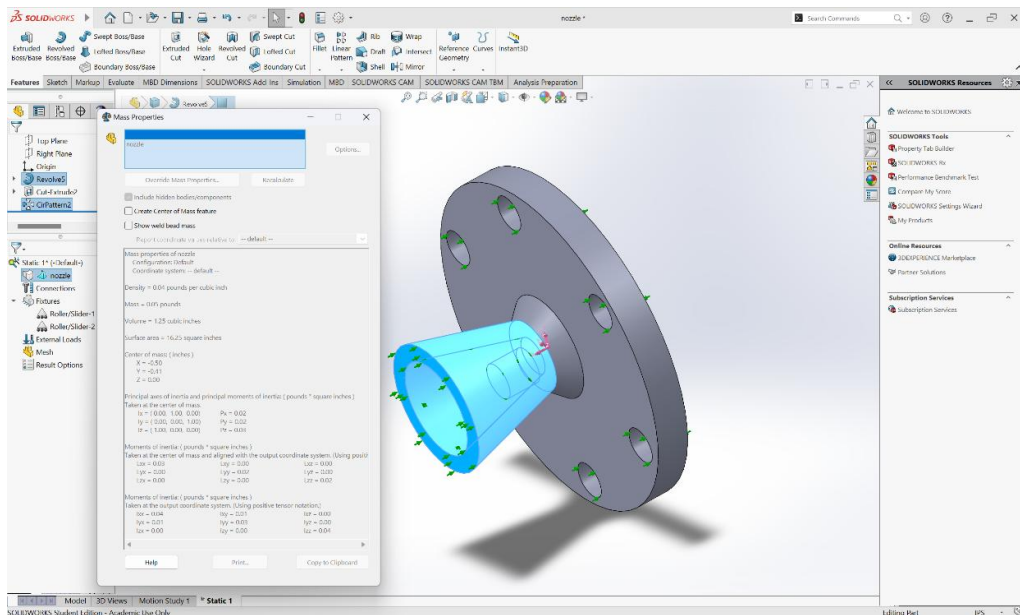
Below, you can see the final product processes for assembling, soldering, wiring and 3D designing.





CAD Design and Engineering Project 3 – Rocket Propulsion Group

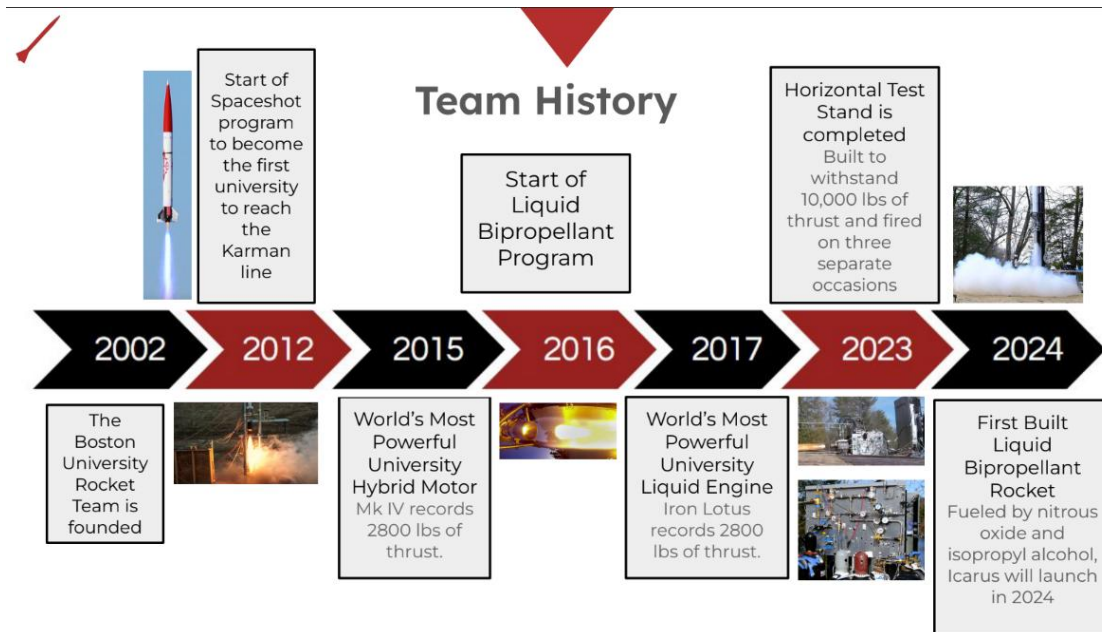
I am involved in a group where the world's most powerful university hybrid motor and liquid engine is developed. I studied on the Hybrid Intro Project where I worked on building and designing a nozzle using Solidworks. While developing the design, I read “Rocket Propulsion Elements” book by George P. Sutton and Oscar Biblarz to make specific adjustments on SolidWorks design. As a result, I printed the design in 3D and tested it. This benefited my skills of learning complex topics by myself and applying my learnings on my CAD designs.



You can see the test and the result below:



I am proud to be involved in a team in which space related projects are accomplished by undergraduate students. You can see the timeline of the projects below:



I also worked on the Icarus Flight Vehicle. In this project, the engine is called Iron Lotus and it produces 2500 lbf. The estimated apogee is 30000 feet and we got the top speed as 1250 mph. We used nitrous oxide as its oxidizer and isopropyl alcohol as its fuel.

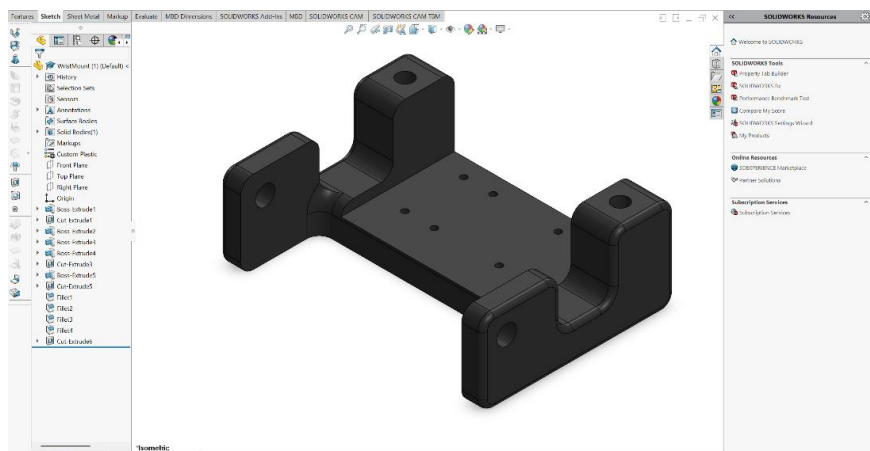


CAD Design and Engineering Project 4 – Mars Rover Club

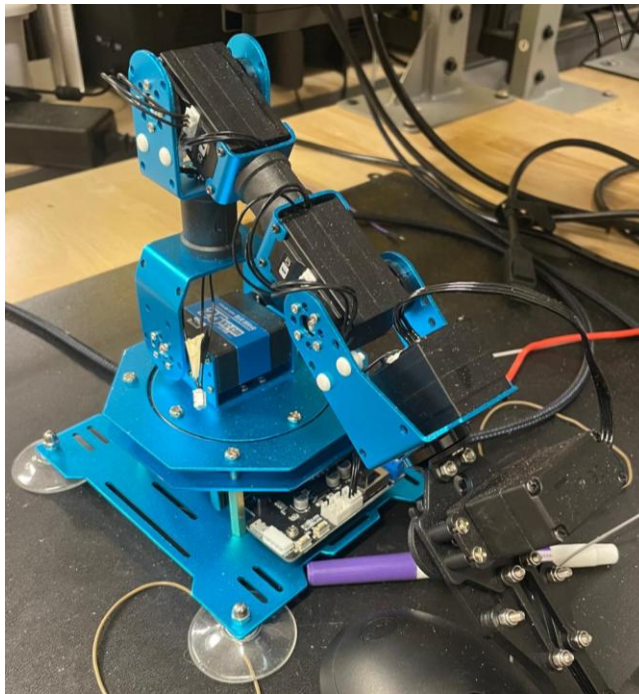
I participated in a group which aims to develop mars rover that is going to be represented in a mars rover competition. I focused on the robotic arm hardware including its material selection, design, testing and construction. Since the design for some components are confidential, I put the final assembled design for robotic arm below.



Below, you can see some of the CAD component designs that I developed for the robotic arm.



Below, you can see the robotic arm for the mars rover.



I also helped to assemble other parts of the mars rover other than the robotic arm. This part forms most of the hardware including the wheels.



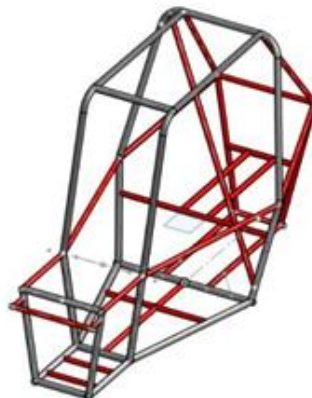
CAD Design and Engineering Project 5 – BAJA Club Off Road Automotive Frame Design

I helped to design and manufacture a competitive off-road vehicle, compliant with the Society of Automotive Engineer's rules to participate in annual SAE events. During this process, I improved knowledge on Solidworks and onshape applications by developing and manufacturing an off-road vehicle. I also gained hands-on skills on welding, design processes and machining.

We worked on improving the design of the off-road automotive represented below according to the changed competition requirements.

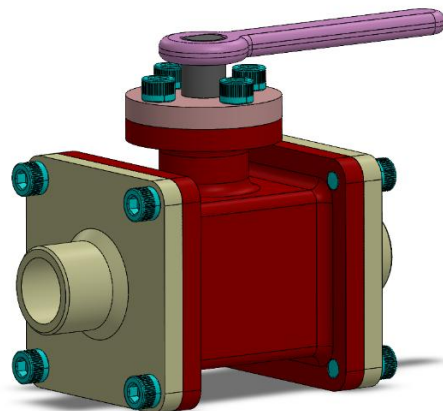
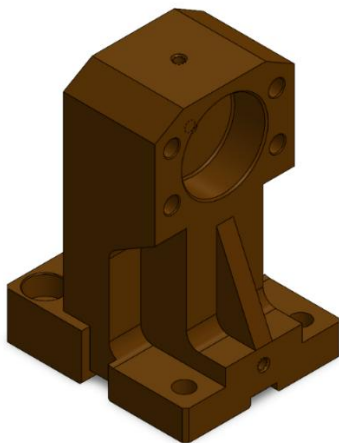
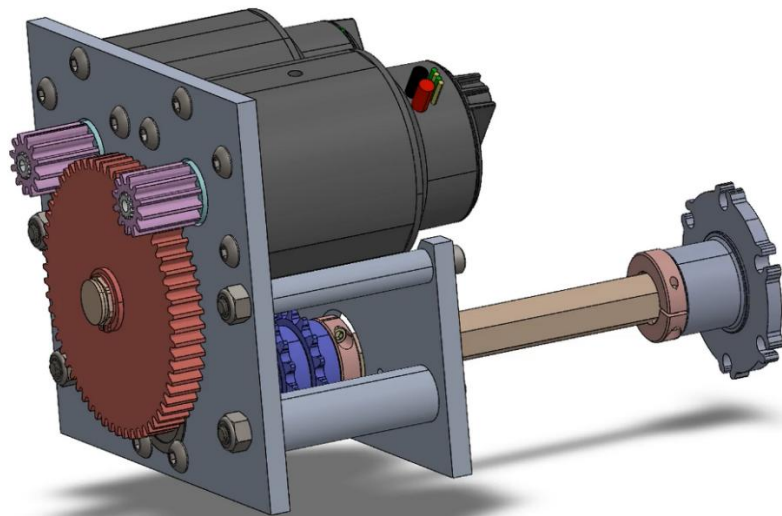


I took role to design lighter and smaller car frame design which is important to increase the speed of the automotive.



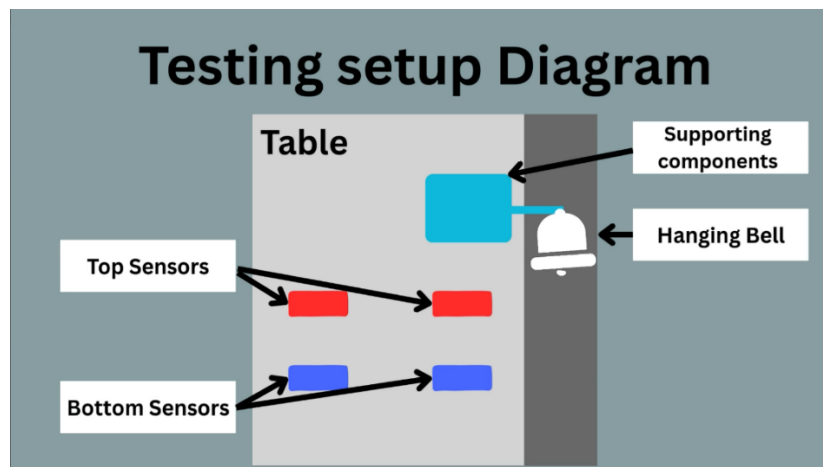
CAD Design and Engineering Project 6 – Additional SolidWorks Designs

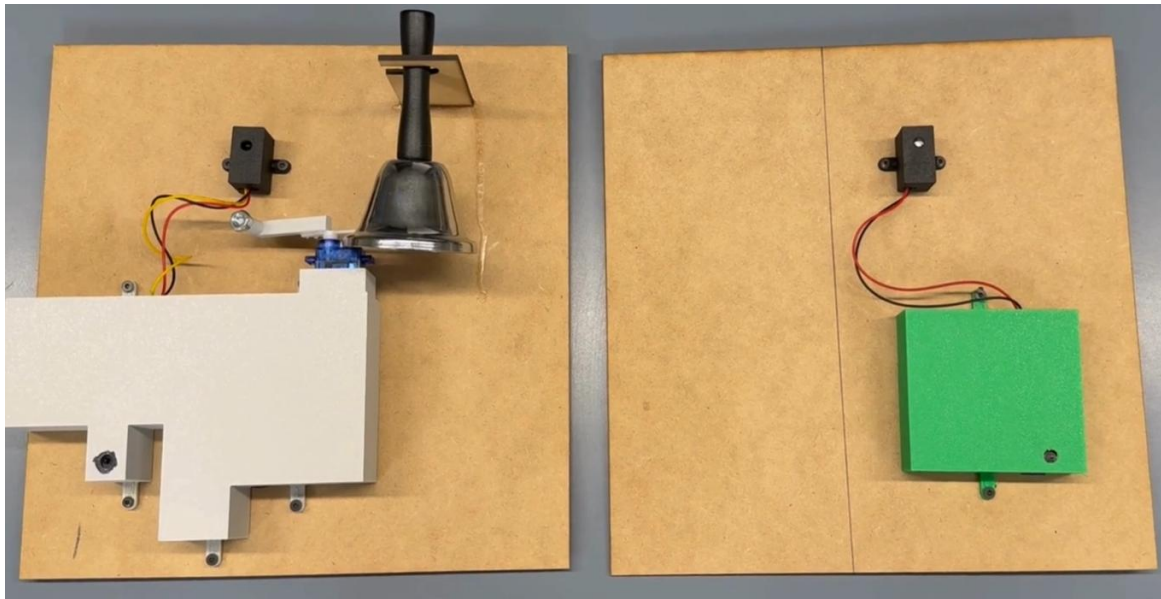
To practice my SolidWorks skills, I created different designs. The designs include gears, video simulations, finite element analysis and component rotations which are important for engineering designs. Also, I acquired their CAD drawings which include the dimensions of the components.



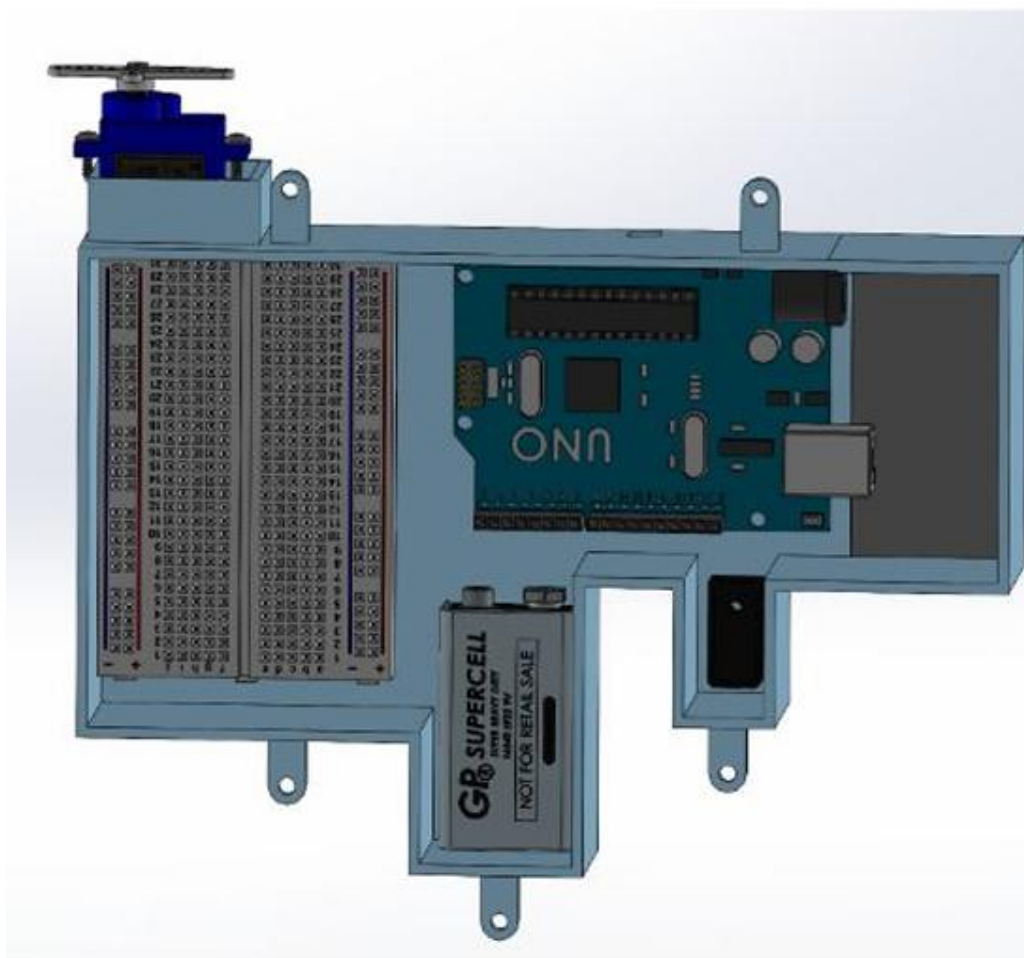
CAD Design and Engineering Project 7 – Retro Passage Bell

The project's goal is to create a retro-style doorbell that distinguishes between humans and pets while incorporating modern functionality. The mechanical chime mechanism is activated by a servo motor and it produces a clear and consistent chime. The placement of IR beam sensors allowed the differentiation of pets and humans and the number of bell rings that inform the user about whether a pet or human passes works accurately. While one part of the IR beam sensor emits a light from one end, the other part of the sensor receives it. When it is interrupted, the sensor is activated, and it leads the servo motor to work. As the servo motor starts to work, the hammer rotates and hits the bell which leads the chime to be heard. The placement of IR beam sensors to different heights allows the differentiation between pets and humans.

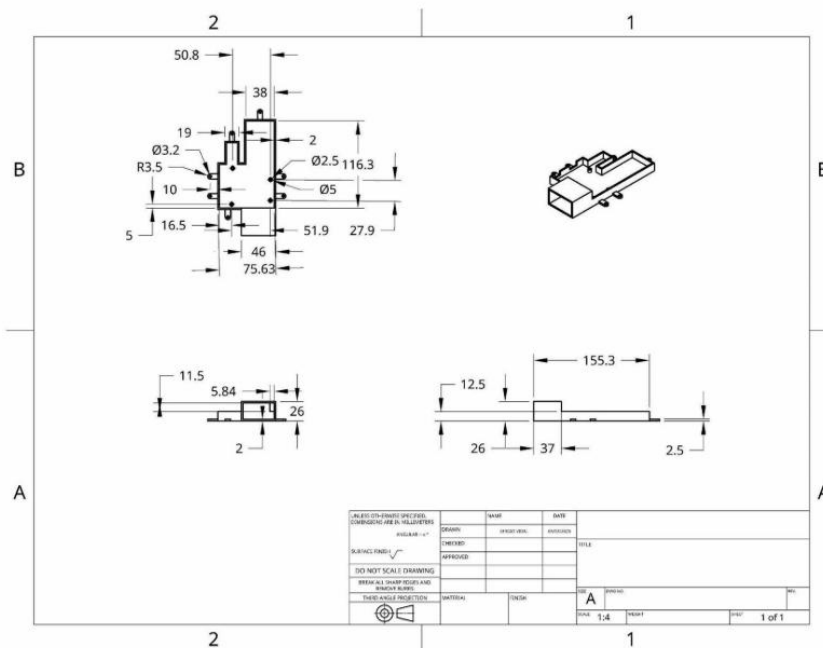
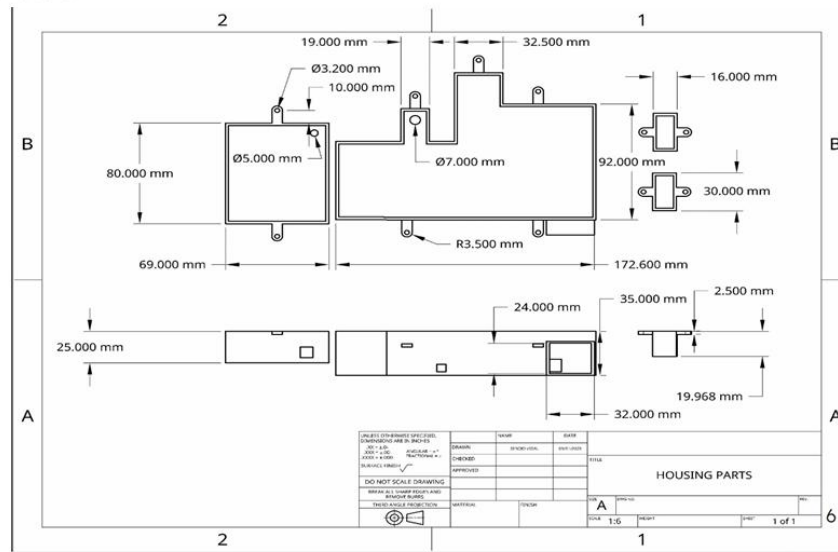




The pattern is protected with a durable housing that ensures long term-reliability in a variety of settings.



CAD Sketch:



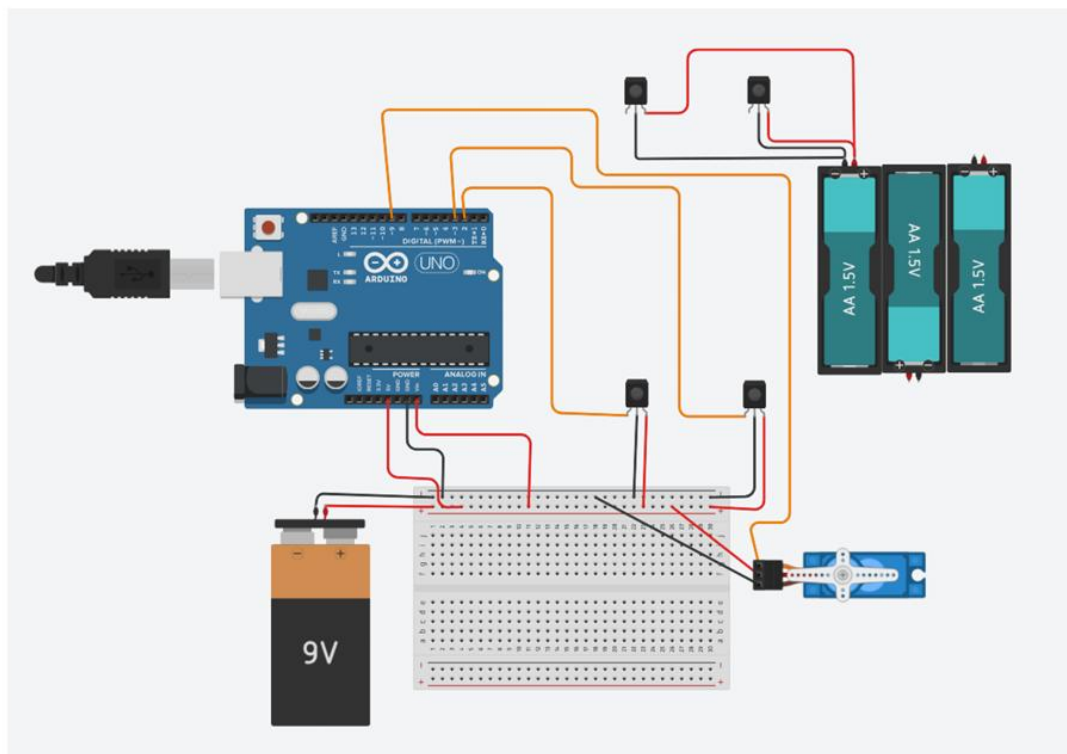
The doorbell was developed to accommodate families with pets and also users seeking a product with a nostalgic feel. To make the project more accessible for different groups, inclusivity was decided as one of the top priorities. Thus, the design is suitable for different groups including people with different ages, skin color, ethnicity and gender. Since IR beam sensors work by emitting and receiving light, it does not make any assumptions about the different characteristics of people. Also, the placement of sensors at different heights make it accessible for people with dwarfism and people using wheelchairs. The project can also be used in different cases like pandemic since it needs no physical interaction.

Below, you can see the important factors that I considered like bill of materials, electronics sketch, code flow chart and various test results while developing the project.

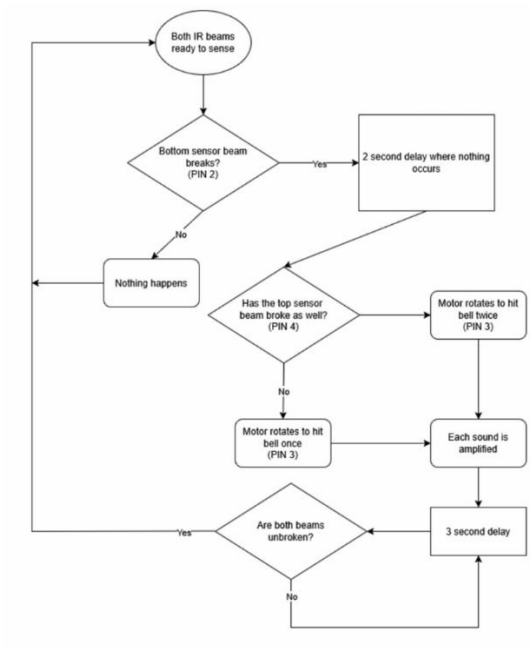
Bill of materials

Component	Units	Price per unit	Total
Arduino Uno	1	\$ 9,59	\$ 9,59
9V battery clip	1	\$ 3,79	\$ 3,79
Breadboard	1	\$ 1,63	\$ 1,63
IR beam sensor	2	\$ 5,95	\$ 11,90
1.5V battery holder	1	\$ 2,25	\$ 2,25
Servo motor	1	\$ 1,80	\$ 1,80
M3 screws	11	\$ 0,02	\$ 0,22
M3 nuts	11	\$ 0,02	\$ 0,22
1.5 V battery	3	\$ 1,25	\$ 3,75
9V battery	1	\$ 2,50	\$ 2,50
ABS	N/A	N/A	\$ 5,00
Bell	1	\$ 6,92	\$ 6,92
		Total	\$ 49,57

Electronics Sketch:

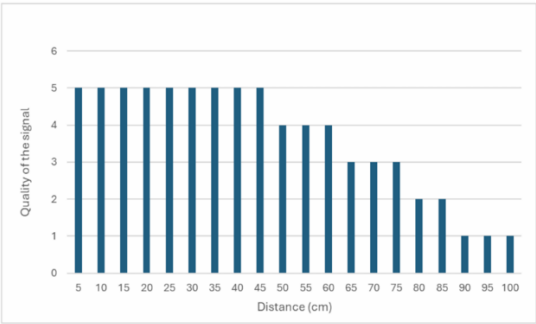


Code Flow Chart:



Distance Test Results

The signal is ranked from 1 (worst) to 5 (best)



Speed Test Results:

The signal is ranked from 1 (worst) to 5 (best) and the speed is ranked from 1 (slow) to 3 (fast)

